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Acetogenic Methane-Carbon Monoxide Comproportionation: An Exergonic but Unobserved Microbial Metabolism

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Key Points:

- Carbon comproportionation reactions are an underexplored class of carbon redox processes relevant to microbial energy conservation
- Methane-carbon monoxide comproportionation to acetate is exergonic across diverse anoxic environments where both substrates co-occur
- Methane-carbon monoxide comproportionation to acetate is also exergonic under plausible early Earth and Mars physicochemical conditions

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Microbial metabolism relies on redox reactions that exploit chemical disequilibria. While aerobic carbon oxidation, carbon fixation, and fermentation are well studied, the broader space of anaerobic carbon redox reactions remains underexplored. In this study, carbon comproportionation, or reverse fermentation, reactions are identified as a previously unrecognized and potentially favorable class of microbial carbon redox transformations. Particular attention is given to the reaction between methane (CH₄) and carbon monoxide (CO) to form acetate, a reaction that has not been evaluated previously despite the widespread occurrence of CH₄ and CO in anoxic systems. Gibbs energies (ΔG_r) for this reaction were calculated across broad ranges of temperature, pH, and dissolved CH₄ and CO concentrations using measured physicochemical data from a wide variety of environmental systems. We show that acetogenic CH₄-CO comproportionation is exergonic in the majority of environments where both substrates were detected. The most favorable energetic conditions occur at high pH, low temperature, and high reactant concentrations, consistent with cool serpentinizing systems. In several settings, the calculated Gibbs energy yields and energy densities overlap or exceed known anaerobic metabolisms involving CH₄, CO, and acetate. These results demonstrate that acetogenic CH₄-CO comproportionation can support microbial energy conservation in a variety of settings. To determine if this metabolism could have operated on early Earth or Mars, modeled fluid compositions show that this reaction is also exergonic under plausible physicochemical regimes. This work broadens the suite of possible microbial energy metabolisms and provides testable criteria for evaluating carbon-based catabolic reactions on Earth and on other planetary bodies.

Plain Language Summary Microorganisms obtain energy by catalyzing chemical reactions in their environment. The energy available from a reaction can be quantified using Gibbs energies of reaction (ΔG_r). When $\Delta G_r < 0$, energy is released that microorganisms can use to build biomass and carry out other activities. In this study, we predicted a new energy-yielding reaction that could potentially support microbial life. In this reaction, methane (CH₄) is oxidized using carbon monoxide (CO) to produce acetate. Using thermodynamic calculations and measured geochemical data from natural environments, we show that this reaction can release usable energy under a wide range of conditions, including those found in continental serpentinizing systems, the deep continental and marine subsurface, and geothermal springs. We also predict that this reaction could support life under plausible early Earth conditions and in modeled Martian fluids. Together, these observations identify the reaction of CH₄ and CO to form acetate as a potentially viable microbial energy source in anoxic environments on Earth and other planetary bodies.

1. Introduction

Over the past half century, first principles thermodynamic calculations (theory) have been utilized to predict microbial catabolic reactions prior to their discovery in nature (empirical evidence). If a redox reaction is sufficiently exergonic (energy-yielding), it can, in principle, support microbial life. This thermodynamics-guided framework has identified catabolic reactions that are now recognized as major contributors to global biogeochemical cycling. Notable examples include anaerobic ammonia oxidation with nitrite or nitrate (anammox) (Broda, 1977; Van de Graaf et al., 1995), which accounts for an estimated 30%–50% of oceanic nitrogen loss (Hutchins & Capone, 2022), and the anaerobic oxidation of methane with sulfate (AOM) (Barnes & Goldberg, 1976; Boetius et al., 2000; Hinrichs et al., 1999; Orphan et al., 2001), which consumes 85–300 Tg of

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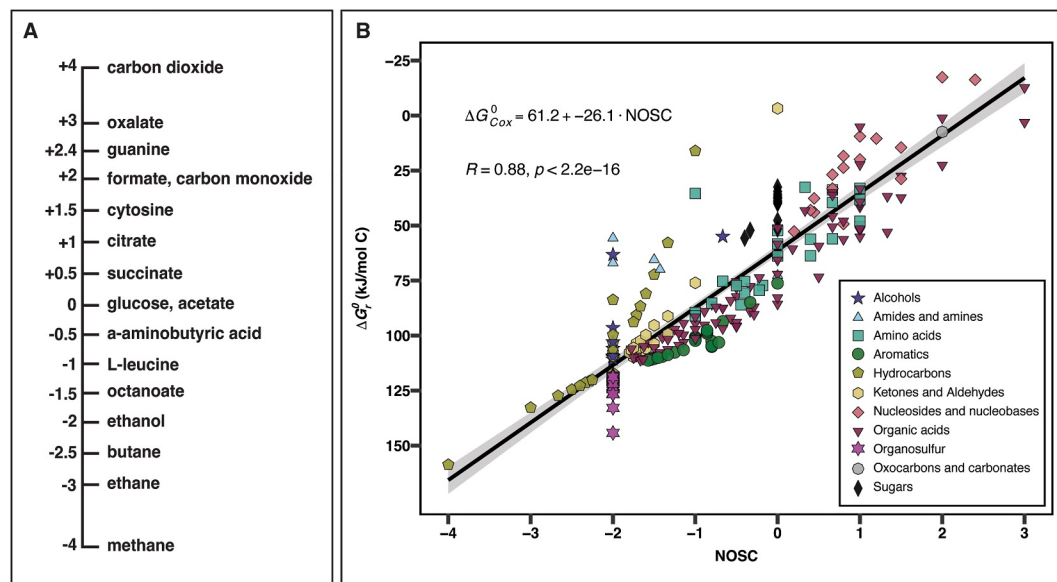


Figure 1. (a) The NOSC of selected carbon compounds. (b) Standard state Gibbs energies, (ΔG_r^0) (kJ/mol carbon), at 25°C calculated for half reactions describing the complete oxidation of each compound to bicarbonate at 25°C and 1 bar, as a function of the nominal oxidation states of carbon, or NOSC. The shapes and colors of the symbols correspond to the indicated classes or organic compounds.

methane annually (Knittel & Boetius, 2009). The prediction of complete ammonia oxidation (comammox) likewise preceded its discovery in nature (Costa et al., 2006; Daims et al., 2015; van Kessel et al., 2015). More recently, thermodynamic predictions have suggested additional “missing” metabolisms, including ammonia-driven methanogenesis, a number of redox reactions among Mn-bearing species, and sulfur comproportionation (Amend et al., 2019, 2020; LaRowe et al., 2021). Indeed, recent experimental work by Aronson and colleagues supports sulfur comproportionation in a strain of *Acidithiobacillus thiooxidans* (Aronson et al., 2025). These studies demonstrate that thermodynamic space still contains plausible yet unrecognized microbial metabolic reactions.

Comproportionation reactions represent one such unexplored class of redox transformations, where two chemical species containing the same element in different oxidation states react to form a product with an intermediate oxidation state. Several known microbial metabolisms follow this pattern, including anammox, which combines nitrite (oxidation state +3) and ammonium (−3) to form N_2 (+0) (Van de Graaf et al., 1995), and sulfur comproportionation, which produces elemental sulfur (+0) from sulfate (+6) and sulfide (−2), and for which a putative microbial candidate has been identified (Amend et al., 2020; Aronson et al., 2025). Comproportionation reactions are in essence reverse disproportionation, or fermentation, reactions, in which a compound of intermediate oxidation state acts simultaneously as an electron donor and acceptor to form two new compounds with divergent oxidation state (e.g., glucose fermentation to ethanol and carbon dioxide: $C_6H_{12}O_6 \rightarrow 2CO_{2(aq)} + 2C_2H_5OH$, where the comproportionation reaction would be the reverse: $2CO_{2(aq)} + 2C_2H_5OH \rightarrow C_6H_{12}O_6$). Although comproportionation is well established for nitrogen and recently for sulfur, analogous reactions involving carbon have received less attention, despite carbon's central role in microbial metabolism (e.g., assimilation, respiration, fermentation, and global biogeochemical cycling).

Carbon exists in oxidation state from −4 (e.g., methane) to +4 (e.g., CO_2), with many intermediates occupying nominal oxidation states of carbon (NOSC) that can serve as products or substrates in redox transformations (Figure 1). Numerous combinations of carbon compounds can theoretically undergo carbon comproportionation reactions, and some of these reactions may be sufficiently exergonic to support microbial energy metabolism. However, the breadth of this reaction space complicates the identification of environmentally relevant and physiologically plausible pathways. To evaluate potential carbon comproportionation reactions, we first surveyed a range of candidate reactions spanning a range of carbon oxidation states using calculations of the standard state Gibbs energy (ΔG_r^0) (Figure 2). This screening step was designed to identify reactions that warrant further

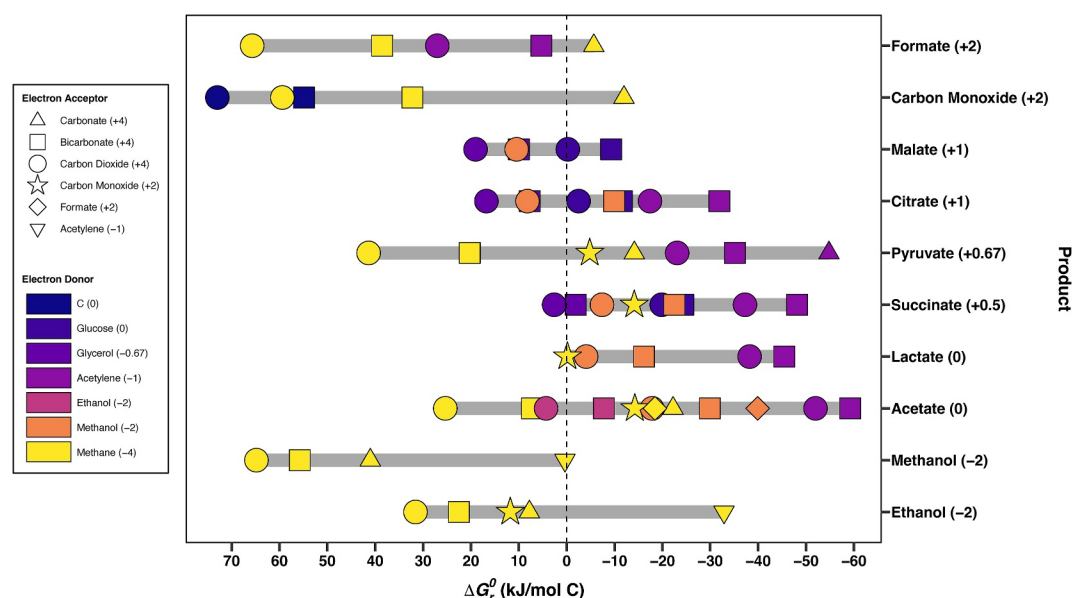
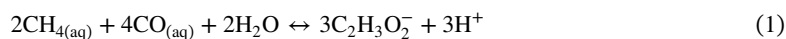


Figure 2. Standard state Gibbs energies, ΔG_r^0 (kJ/mol carbon), at 25°C for 65 different carbon comproporation reactions. Each point represents a unique donor-acceptor-product combination. Point color denotes the electron donor, point shape denotes the electron acceptor, and the vertical position corresponds to the product compound (listed along the right side of the plot). For example, the left-most symbol in the top line corresponds to the reaction of $\text{CH}_{4(\text{aq})} + 3\text{CO}_{2(\text{aq})} + 2\text{H}_2\text{O} \rightarrow 4\text{HCO}_2^- + 4\text{H}^+$. The numbers in parentheses next to each product species refer to the nominal oxidation states of carbon (NOSC) in that compound.

consideration under environmentally realistic conditions. From this broader reaction space, we focus on a single carbon comproporation reaction of particular environmental relevance: the reaction of methane (CH_4 ; NOSC = -4) and carbon monoxide (CO ; NOSC = +2) to form acetate (NOSC = 0) (Reaction 1). This reaction links two C1 carbon compounds that co-occur and are widely cycled in many anoxic environments and produces acetate, a central metabolite in anaerobic carbon metabolism, including acetogenic carbon fixation and fermentation. More broadly, reactions involving carbon monoxide are generally exergonic, yet they remain comparatively underexplored relative to other carbon catabolic reactions. Importantly, paired measurements of CO and CH_4 are available in a variety of natural sites, enabling quantitative evaluation of this reaction across diverse geochemical settings. This data availability allows acetogenic CH_4 - CO comproporation to be assessed not only as a theoretical possibility but also within the geochemical contexts where it may realistically occur. As such, this reaction represents a plausible carbon comproporation catabolism that could have operated under early Earth conditions, may be viable in extraterrestrial environments, and could be occurring today in certain extreme environments.



Here, we evaluate the thermodynamic feasibility of acetogenic CH_4 - CO comproporation as a potential mode of microbial energy conservation following the approach of Amend et al. (2020) and Aronson et al. (2025) for S-comproporating metabolisms. We calculate Gibbs energies (ΔG_r) across wide ranges of temperature, pH, and ionic composition. These calculations incorporate measured physicochemical data from several anoxic environment types, including serpentinizing systems, deep continental and marine subsurface fluids and sediments, and geothermal springs, as well as modeled early Earth and Martian conditions. By performing thermodynamic calculations under a variety of actual geochemical environments, this study delineates the natural conditions under which carbon comproporation reactions may be energetically viable and establishes a framework for assessing their potential role in anoxic ecosystems on Earth, in Earth's past, and on other planetary bodies.

2. Methods

The nominal oxidation state of carbon (NOSC) was calculated for 253 organic compounds from the CHNOSZ database (Dick, 2019) using,

$$\text{NOSC} = -\left(\frac{-Z + 4a + b - 3c - 2d + 5e - 2f}{a}\right) + 4 \quad (2)$$

where Z is the net charge of the compound, and a, b, c, d, e , and f correspond to the stoichiometric numbers of the elements C, H, N, O, P, and S in each compound (LaRowe & Van Cappellen, 2011). A half reaction describing complete oxidation to bicarbonate was written for each compound (Table S1 in Supporting Information S2), and the standard state Gibbs energies of carbon oxidation, ΔG_{Cox}^0 , were calculated for each at 25°C and 1 bar with the revised Helgeson-Kirkham-Flowers (HKF) equations of state (Helgeson et al., 1981; Shock & Helgeson, 1988; Tanger & Helgeson, 1988) with the “subct” command in the R software package CHNOSZ v1.4.1 (Dick, 2019). Thermodynamic data implemented in CHNOSZ are sourced from the OrganoBioGeoTherm database; original data sources and references are provided in Methods in Supporting Information S1.

The standard state Gibbs energy of reaction, ΔG_r^0 , was calculated for 65 carbon comproportionation reactions as described above (Table S2 in Supporting Information S2). Electron donors (and their oxidation states) included methane (−4), ethanol (−2), methanol (−2), acetylene (−1), glycerol (−0.67), glucose (0), and elemental carbon as graphite (0). These substrates include widely utilized biological carbon sources (e.g., methane, glucose, methanol, ethanol) and compounds involved in previously proposed or demonstrated comproportionation reactions (Kremp et al., 2018; Lahijani et al., 2015; Lever, 2012; Litty et al., 2022; Steiger et al., 2017). Electron acceptors included CO (+2), formate (+2), and the dissolved inorganic carbon (DIC) $\text{CO}_2/\text{HCO}_3^-/\text{CO}_3^{2-}$ (+4), in addition to acetylene. Ten product compounds were chosen from physiologically relevant metabolic intermediates: formate (+2), carbon monoxide (+2), malate (+1), citrate (+1), pyruvate (+0.67), succinate (+0.5), lactate (0), acetate (0), methanol (−2), and ethanol (−2). These compounds are common in microbial fermentation reactions, carbon fixation pathways (e.g., Wood-Ljungdhal, reverse TCA), and organic acid metabolism (Ragsdale & Pierce, 2008).

Gibbs energies of reaction, ΔG_r , were calculated for Reaction 1 with Equation 3:

$$\Delta G_r = \Delta G_r^0 + RT \ln Q \quad (3)$$

where ΔG_r^0 is the standard state Gibbs energy, R denotes the universal gas constant, T represents to the temperature in Kelvin, and Q_r refers to the reaction quotient. Values of Q_r were calculated using Equation 4:

$$Q_r = \prod_i a_i^{v_{i,r}} \quad (4)$$

where a_i corresponds to the activity of the i th species raised to its stoichiometric coefficient, $v_{i,r}$, in the r th reaction, which is positive for products and negative for reactants. Activities were calculated with Equation 5.

$$a_i = \gamma_i \left(\frac{C_i}{C_i^0} \right) \quad (5)$$

where γ_i and C_i represent the activity coefficient and the measured concentration of the i th species, respectively. C_i^0 is the concentration of the i th species under standard state conditions, which is equal to one molal referenced to infinite dilution. Measured concentrations were typically reported in molarity; for activity calculations, these values were treated as equivalent to molality.

ΔG_r was calculated for Reaction 1 at 1 bar across a range of pH (0–14) and temperature (0–100°C) with the activities of products and reactants (except for H^+) set at environmentally relevant activities. The activities of methane (CH_4), carbon monoxide (CO), and acetate were set at 10^{-5} ($\approx 10 \mu\text{M}$), 10^{-9} ($\approx 1 \text{ nM}$), and 10^6 ($\approx 1 \mu\text{M}$), respectively.

Concentrations of CH₄ and CO from a variety of environments were taken from a number of published sources (Boulart et al., 2013; Canovas et al., 2017; Cardace et al., 2015; Cook, Blank, Rietze, et al., 2021; Cook, Blank, Suzuki, et al., 2021; Crespo-Medina et al., 2014; Eickenbusch et al., 2019; Gihring et al., 2006; Howells et al., 2025; Kieft et al., 2005; Lau et al., 2016; Lu et al., 2020; Magnabosco et al., 2016; Mara et al., 2023; Momper et al., 2023; Morrill et al., 2014; Moser et al., 2005; Nyssönen et al., 2012; Onstott et al., 2009; Osburn et al., 2014, 2019; Qi et al., 2025; Robare, 2025; Robare et al., 2026; Sabuda, 2017; Sabuda et al., 2021; Twing et al., 2017) (Table S3 in Supporting Information S2). For samples in which CO and CH₄ were reported as below detection limits, concentrations were assigned as one order of magnitude lower than the analytical detection limit for the compound specified in each study. Within the serpentinite mud samples (Eickenbusch et al., 2019), CH₄ was below detection limits (<0.01 μM) in 10 out of 67 samples, and CO was below detection limits (<0.2 μM) in 16 out of 67 samples. These samples were assigned concentrations of 10⁻⁸ M CH₄ and 2 × 10⁻⁸ M CO. Within serpentinizing spring samples, CO was below detection limits (<10 μM) in 10 out of 88 samples. These samples were assigned concentrations of 10⁻⁷ M CO. This approach avoids overestimating reactant activities, which would otherwise bias the ΔG_r estimates toward more negative values. The ΔG_r values calculated from these substituted activities should therefore be interpreted as conservative estimates of the potential energy yield of Reaction 1 under low-CO and low-CH₄ conditions. For samples in which acetate was not reported, the acetate concentration was set to 1 μM. This value falls within the range of acetate concentrations measured across many anoxic environments and provides a reasonable estimate for ΔG_r calculations. Activities were calculated from the in situ or estimated chemical compositions of each sample using the “ae.speciate” command from the aqueous speciation package AqEquil v0.18.0 (Boyer et al., 2024) (Table S4 in Supporting Information S2).

Each ΔG_r⁰ was calculated as described above for 439 natural samples at the reported in situ temperatures and pressures. Water pressure at the seafloor was calculated for marine subsurface sediment samples (Methods in Supporting Information S1). Geochemical data for hypothetical Mars fluids were estimated using data from Marlow et al. (2014) and Rucker et al. (2023), with atmospheric CO concentrations set at either 1% or 50% (Koyama et al., 2024), for a total of 22 samples. Henry's law was used to calculate the concentration of aqueous CO from gaseous CO at 7°C, 9°C, 14°C, 15°C, 24°C, 48°C, 65°C, and 83°C. Henry's law constants (mol L⁻¹ atm⁻¹) for CO were calculated at each temperature using the equilibrium constants of the gas dissolution reaction for each compound at 2.2 bar. Geochemical data for a hypothetical early Earth ocean were estimated by determining the equilibrium activities of aqueous CH₄ corresponding to partial pressures ranging from 5 × 10⁻³ to 1.8 × 10⁻² bar and aqueous CO corresponding to partial pressures ranging from 10⁻⁵ to 10⁻² bar at temperatures from 0 to 40°C, and pH 7 (Halevy & Bachan, 2017), using the ionic strength of the modern ocean without sulfate, for a total of 164 conditions.

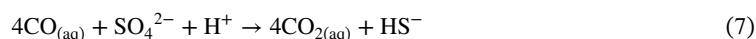
Gibbs energies, ΔG_r, were calculated for Reaction 1 as described above for 625 samples (439 natural samples and 186 modeled conditions) as described above. Negative values of ΔG_r indicate exergonic (energy-yielding) reactions whereas positive values indicate endergonic (energy-consuming) reactions. To compare Reaction 1 with known anaerobic microbial metabolic reactions that consume CO or CH₄ or produce acetate, ΔG_r was calculated for six additional reactions for each sample under the geochemical parameters described above. These reactions include the water-gas shift reaction (Reaction 6), CO oxidation coupled to sulfate reduction (Reaction 7), methanogenic CO disproportionation (Reaction 8), acetogenic CO disproportionation (Reaction 9), methane oxidation coupled to sulfate reduction (AOM; Reaction 10), and homoacetogenesis (Reaction 11). For sites where sulfate, sulfide, or hydrogen activities were absent, activities of 10⁻⁶ were assumed.

The reactions evaluated are:

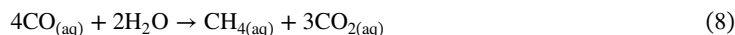
Water gas shift



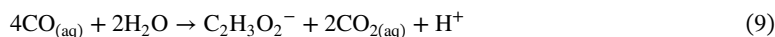
CO oxidation coupled to sulfate reduction



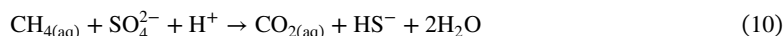
Methanogenic CO disproportionation



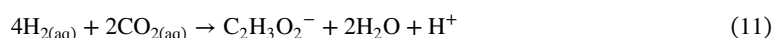
Acetogenic CO disproportionation:



Anaerobic oxidation of methane (AOM)



Homoacetogenesis



To account for differences in substrate availability among reactions and environments, energy densities (E_r ; kJ/kg H_2O) were calculated to weight the Gibbs energy by the availability of the limiting reactant:

$$E_r = \frac{\Delta G_r [a_{\text{lim}}]}{\nu_{\text{lim}}} \quad (12)$$

where a_{lim} is the activity of the limiting reactant and ν_{lim} is its stoichiometric coefficient. Energy densities represent the total accessible energy available per kilogram of water under the specified geochemical conditions. For quantitative comparisons among reactions, Reaction 1 was treated as a reference reaction, and pairwise differences in energy density were calculated for each sample. Differences in energy density were evaluated using the log-transformed metric $\Delta \log E$, defined for samples in which both reactions were exergonic as:

$$\Delta \log E = \log_{10}(-E_r) - \log_{10}(-E_1) \quad (13)$$

where E_r and E_1 are the energy densities of the comparison reaction and Reaction 1, respectively. $\Delta \log E$ was calculated only for samples in which both reactions are exergonic, as energy densities for endergonic reactions are positive and therefore yield non-real values under the $\log_{10}(-E_r)$ transformation. Negative values of $\Delta \log E$ indicate higher energy density for Reaction 1, whereas positive values indicate higher energy density for the comparison reaction. For each reaction, the null hypothesis that the median $\Delta \log E$ equals zero was evaluated using a Holm-corrected Wilcoxon signed-rank test. Statistical analyses were conducted across the 439 natural samples only. Statistical analyses were performed using R version 4.5.1. All data and R scripts are available at <https://doi.org/10.5281/zenodo.18158636>.

3. Results

3.1. Thermodynamic Potential for Carbon Comproportionation Reactions

Examples of NOSC values for common compounds are shown for reference in (Figure 1a). Standard state Gibbs energies of carbon oxidation (ΔG_{Cox}^0) for 253 carbon-containing compounds are shown in Figure 1b. These compounds span the full range of NOSC values and include alcohols, amides and amines, amino acids, aromatics, hydrocarbons, ketones and aldehydes, nucleosides and nucleobases, organic acids, organosulfur compounds, sugars, oxocarbons, and carbonates. Consistent with earlier work (LaRowe & Van Cappellen, 2011), ΔG_{Cox}^0 shows an inverse correlation with NOSC ($R^2 = 0.85$, $p < 2.2 \times 10^{-16}$). Compounds with higher NOSC values exhibited lower half reaction standard state Gibbs energies (Figure 1b).

We present the standard state Gibbs energies (ΔG_r^0 , at 25°C) for 65 carbon comproportionation reactions relevant to microbial carbon transformations (Figure 2). These reactions span a broad range of NOSC values. Of these the oxidation of acetylene with bicarbonate to produce acetate had the most negative value of ΔG_r^0 (−59 kJ/mol C), whereas the oxidation of elemental carbon with CO_2 to form CO yielded the most positive value (+73 kJ/mol C).

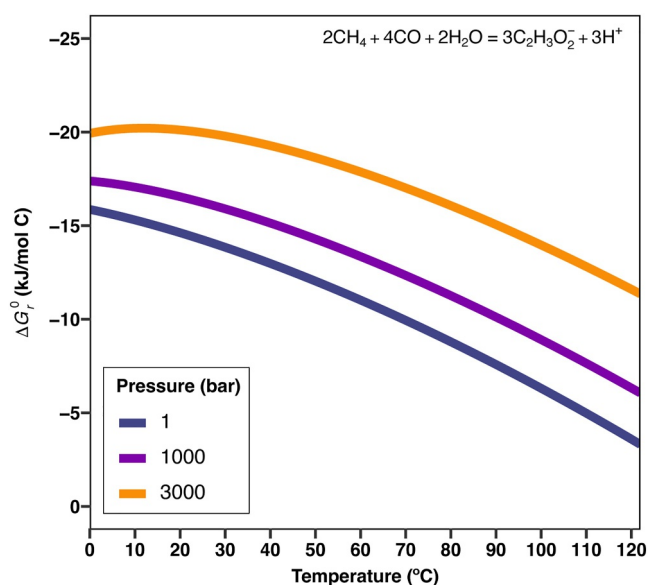


Figure 3. Standard state Gibbs energy, ΔG_r^0 (kJ/mol carbon), for Reaction 1 as a function of temperature. For temperatures $\geq 100^\circ\text{C}$, standard state Gibbs energies were calculated at the saturation vapor pressure of water for each temperature.

Reactions forming pyruvate exhibited the widest range in ΔG_r^0 . For example, the reaction of methane and CO_2 has a standard state Gibbs energy of +41 kJ/mol C, while the acetylene- CO_3^{2-} pair yielded −55 kJ/mol C. By contrast, reactions forming malate showed the narrowest range of ΔG_r^0 , from +19 kJ/mol C for glycerol- CO_2 to +0.4 kJ/mol C for methane-acetylene. Many acetylene-based reactions yielded negative values of ΔG_r^0 (−17 to −59 kJ/mol C).

The standard state Gibbs energies of Reaction 1 (ΔG_r^0) at 1, 1,000, and 3,000 bar are shown as a function of temperature in Figure 3. Values of ΔG_r^0 were negative from 0 to 122°C and became more negative with increasing temperatures, ranging from −15 kJ/mol C at 0°C and −3 kJ/mol at 122°C . At each temperature, values of ΔG_r^0 became more negative with increasing pressure. At 0°C , ΔG_r^0 decreased from −15 kJ/mol C at 1 bar to −20 kJ/mol C at 3,000 bar, and at 122°C ΔG_r^0 decreased from −3 kJ mol $^{-1}$ at 1 bar to −11 kJ mol $^{-1}$ at 3,000 bar. When overall Gibbs energies (ΔG_r) were evaluated at environmentally realistic activities of CH_4 (10 μM), CO (1 nM), and acetate (1 μM), the reaction remained exergonic across broad ranges of pH and temperature (Figure 4). Under these conditions, values of ΔG_r approached −60 kJ/mol C at high pH and low temperature. The strong pH dependence reflects the consumption of three protons in Reaction 1; temperature has a secondary influence on energy yield.

While not an exhaustive list of data sets, we found that CO and CH_4 occurred in 25 geochemical data sets that represented 439 natural samples, together with 28 modeled fluids representing early Earth oceans and Martian environments, for a total of 467 scenarios. The energetics of Reaction 1 across all 467 geochemical scenarios evaluated are plotted in Figure 5. Across this combined data set, ΔG_r values normalized by moles of carbon spanned from −42 to +21 kJ/mol C, with 372 samples revealing negative Gibbs energies (80% of sites). Among the natural samples, pH ranged from 1.8 in springs from Yellowstone National Park to 13.1 at serpentinizing springs at The Tablelands. Temperature ranged from 2°C in serpentinite muds from the Mariana Forearc to 96°C in the deep subsurface sediment samples from Guaymas Basin. The deep continental subsurface samples displayed the widest range of CO measurements, with the lowest and highest reported CO concentrations in the entire data set (0.06 nM at DeMMO to 680 μM at Kloof Mine). The lowest reported CH_4 measurement was from DeMMO (0.08 nM) and the highest was from Guaymas Basin (130 mM). To identify the geochemical variables that exert the strongest influence on ΔG_r , correlations were evaluated using only the natural samples ($n = 439$). Across these samples, pH exhibited a strong linear association with ΔG_r ($R^2 = 0.86$, $p = 1.96 \times 10^{-179}$), consistent with the stoichiometric consumption of three protons in Reaction 1 (Figure 6a). However, this relationship varies among environments, with strong correlations in marine sediments ($R^2 = 0.98$), geothermal springs ($R^2 = 0.76$), and serpentinizing springs ($R^2 = 0.78$), and weaker or non-significant correlations in other settings such as the marine subsurface ($R^2 = 0.09$). Temperature also exerted a significant, but secondary association (Figure 6b) ($R^2 = 0.558$, $p = 1.63 \times 10^{-79}$), reflecting the temperature dependence of ΔG_r^0 (Figure 3).

3.2. Modeled Martian and Early Earth Fluids

Modeled fluid composition for early Earth and Mars spanned a wide energetic range for Reaction 1, and was exergonic across all 186 scenarios evaluated, with ΔG_r values spanning from −8 to −43 kJ/mol C (Figure 5). Under 22 modeled Martian fluid scenarios, including Eridania Basin hydrothermal fluid models (Rucker et al., 2023) and the Mars geochemical models from Marlow et al. (2014), under low- and high- CO atmospheric conditions, ΔG_r ranged from −8 to −43 kJ/mol C. Early Earth ocean compositions (164 modeled scenarios) yielded negative ΔG_r values. The four atmospheric endmembers (CH_4 partial pressures of 5×10^{-3} or 1.8×10^{-2} bar and CO partial pressures of 10^{-5} or 10^{-2} bar) produced dissolved CH_4 concentrations of 4.85–38.5 μM and dissolved CO concentrations of 8.2 nM–16.9 μM over a temperature range of 0– 40°C . Under these conditions, ΔG_r spanned from −9 to −25 kJ/mol C. Six representative points (high or low gas partial

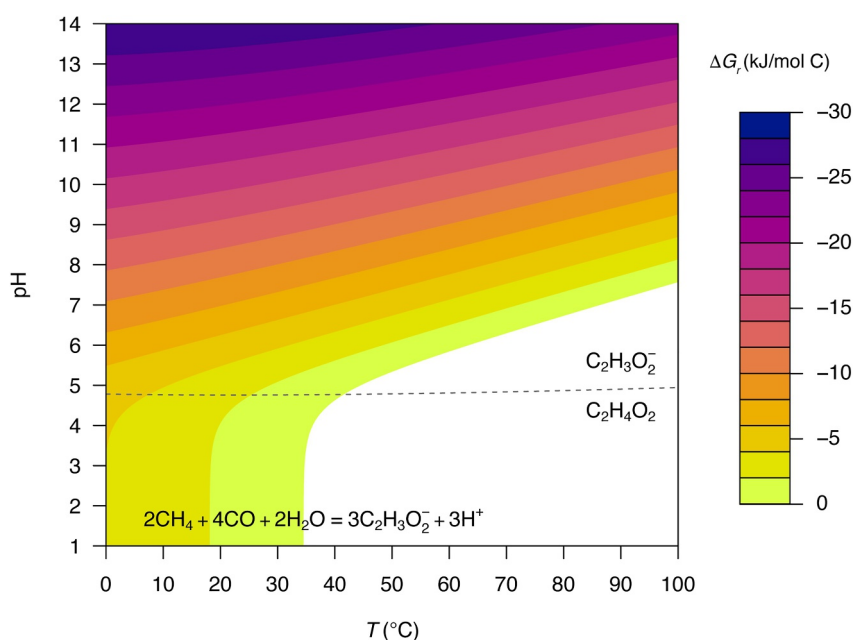


Figure 4. Values of the overall Gibbs energy of reaction, ΔG_r (kJ/mol carbon), for Reaction 1 as a function of temperature and pH. Activities of acetate across the temperature and pH space represented here were calculated with equilibrium speciation among CH_3COO^- and CH_3COOH given a total acetate activity of 10^{-6} . Activities of aqueous CH_4 and CO were set at 10^{-5} and 10^{-9} , respectively.

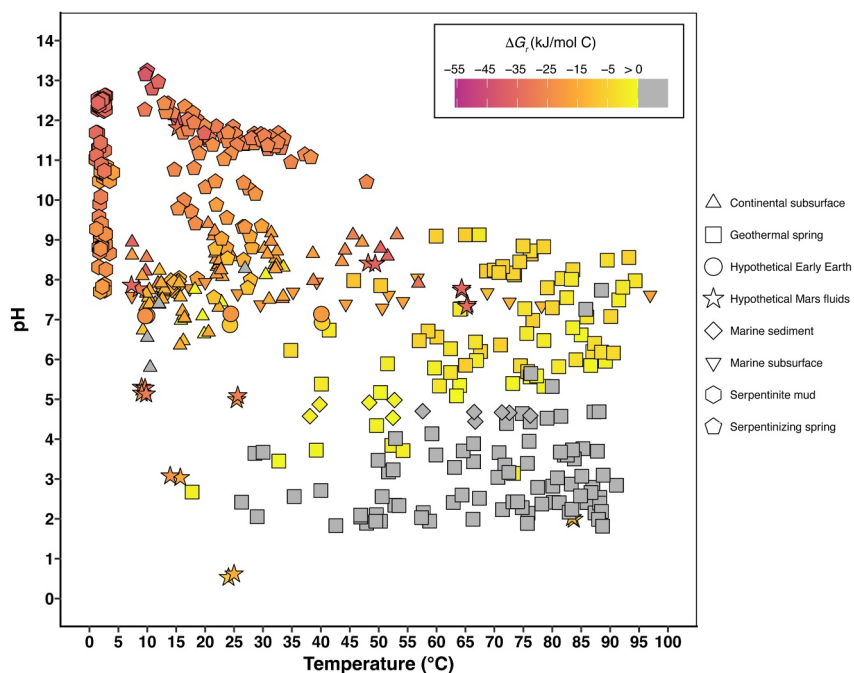


Figure 5. Values of the overall Gibbs energy of reaction, ΔG_r (kJ/mol carbon) for Reaction 1. Values of ΔG_r are shown as a function of temperature and pH for 467 combinations of temperature, pH, pressure, and activities of CH_4 , CO , and acetate representing samples from seven anoxic environment types, along with modeled early Earth ocean and Martian fluids. Temperature, pH, pressure and activities of CH_4 , CO , and acetate were taken directly from in situ geochemical data from seven environment types, represented as the shape of the points. Point fill indicates the magnitude of ΔG_r (kJ/mol C); gray points denote samples where $\Delta G_r > 0$.

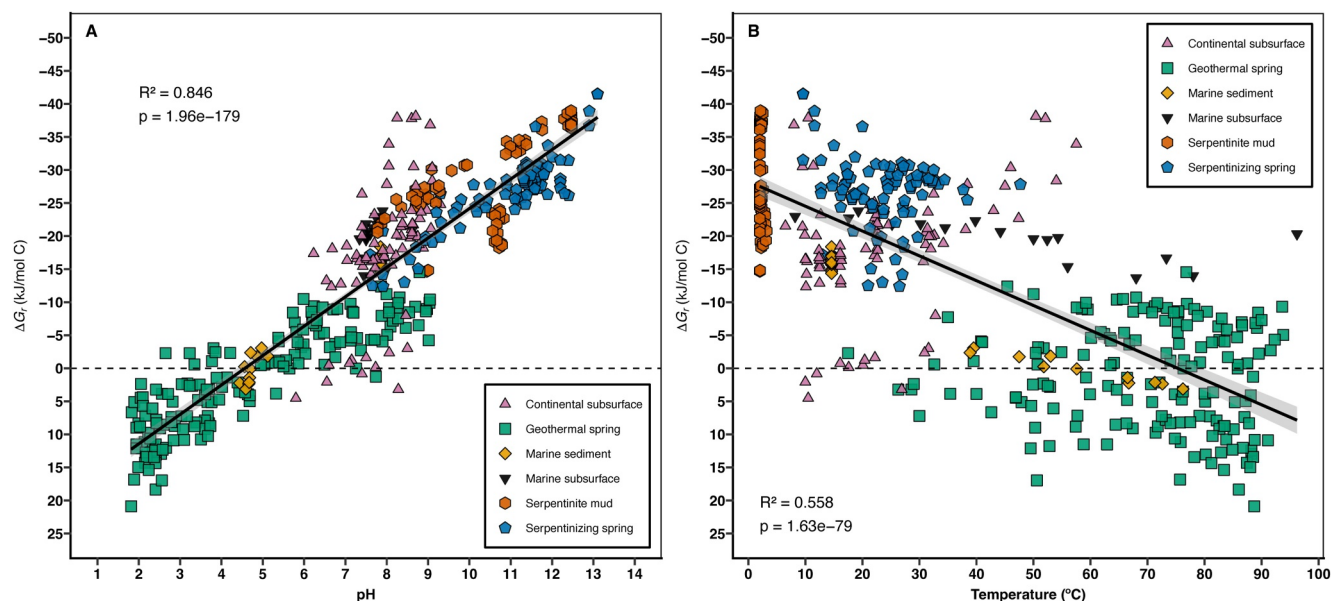


Figure 6. Values of the overall Gibbs energy of reaction, ΔG_r (kJ/mol carbon), for Reaction 1 ($2\text{CH}_4(\text{aq}) + 4\text{CO}(\text{aq}) + 2\text{H}_2\text{O} \rightarrow 3\text{C}_2\text{H}_3\text{O}_2^- + 3\text{H}^+$) represented as a function of (a) pH and (b) temperature ($^{\circ}\text{C}$) from 439 samples from natural anoxic environments. Temperature, pH, and activities of CH_4 , CO , and acetate were sourced from in situ geochemical data from six environment types, represented as the shape of the points.

pressures, at 10, 25, and 40°C), selected to span end-member conditions and an intermediate temperature, are shown in Figure 5 to illustrate the full range of predicted ΔG_r .

3.3. Energy Yields of Other Reactions Involving CO , CH_4 , and Acetate

ΔG_r was calculated for six additional reactions (Reactions 6 through 11) representing known metabolisms that either produce acetate or consume CO and CH_4 under the geochemical conditions of the 439 natural samples (Figure 7). Energy yields were normalized by moles of carbon to facilitate comparisons between samples. Reaction 1 yielded ΔG_r values between +21 to -42 kJ/mol C and was exergonic in 78% of samples. Reaction 6 (water-gas shift reaction) spanned $+0.64$ to -82 kJ/mol C and was exergonic in all natural samples except for one. Reaction 7 (CO oxidation coupled to sulfate reduction) was exergonic in all natural samples and yielded -27 to -109 kJ/mol C. Reaction 8 (methanogenic CO disproportionation) also produced negative values of ΔG_r in all natural samples, ranging from -18 to -98 kJ/mol C. Reaction 9 (acetogenic CO disproportionation) yielded -3 to -86 kJ/mol C and was exergonic in all natural samples. Reaction 10 (methane oxidation coupled to sulfate reduction) yielded $+3$ to -103 kJ/mol C and was exergonic in all but two natural samples. Reaction 11 (homoacetogenesis) was exergonic in 51% of natural samples and yielded ΔG_r values between $+60$ and -82 kJ/mol C.

Energy densities (E_r) were calculated from ΔG_r to represent the amount of energy available in a kilogram of water, normalized to the activity of the limiting reactant (Figure 8). Because energy densities span several orders of magnitude across reactions and environments, values corresponding to exergonic reactions ($\Delta G_r < 0$) are plotted in Figure 8 as $\log_{10}(-E_r)$ to improve visual resolution; all numerical values reported in the text refer to untransformed values of E_r . Across the natural samples, Reaction 1 yielded energy densities ranging from $+2.39 \times 10^{-6}$ kJ/kg H_2O to -3.46×10^{-2} kJ/kg H_2O . Reaction 6 spanned -7.70×10^{-11} to -3.34×10^{-2} kJ/kg H_2O . Reaction 7 ranged from -3.46×10^{-11} to -5.96×10^{-5} kJ/kg H_2O . Reaction 8 yielded -1.92×10^{-9} to -4.99×10^{-2} kJ/kg H_2O . Reaction 9 had energy densities between -3.84×10^{-10} and -4.48×10^{-2} kJ/kg H_2O . Reaction 10 spanned from $+4.98 \times 10^{-8}$ to -6.52×10^{-4} kJ/kg H_2O . Reaction 11 yielded energy densities from $+1.21 \times 10^{-5}$ to -1.65×10^{-4} kJ/kg H_2O .

To evaluate relative energetic favorability across reactions, pairwise differences in the Gibbs energy and log-transformed energy density ($\Delta \log E$) were calculated for each sample using Reaction 1 as a reference (Table 1). Because $\Delta \log E$ is expressed on a $\log_{10}$ scale, a difference of 1 unit corresponds to a 10-fold difference in energy

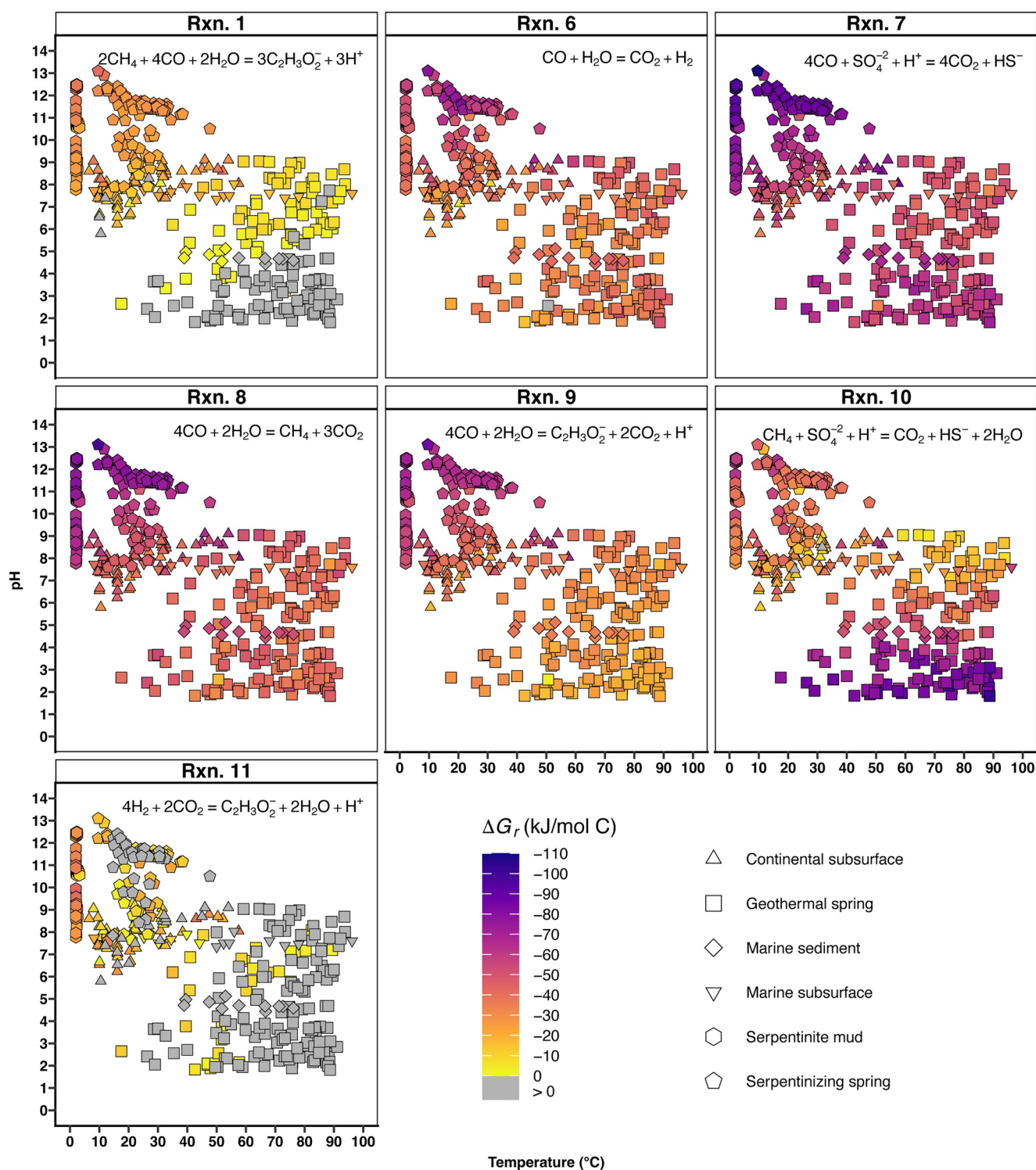


Figure 7. Values of the overall Gibbs energy of reaction, ΔG_r , normalized per mole of carbon for Reaction 1 and reference reactions (Reactions 6–11). Values of ΔG_r are shown as a function of temperature and pH for 439 combinations of temperature, pH, pressure, and activities of CH_4 , CO , acetate, sulfide, sulfide, H_2 , and CO_2 representing natural samples from seven anoxic environment types. Temperature, pH, pressure and activities were taken from speciated in situ geochemical data from seven environment types, represented as the shape of the points. Point fill indicates the magnitude of ΔG_r (kJ/mol C); gray points denote samples where $\Delta G_r > 0$.

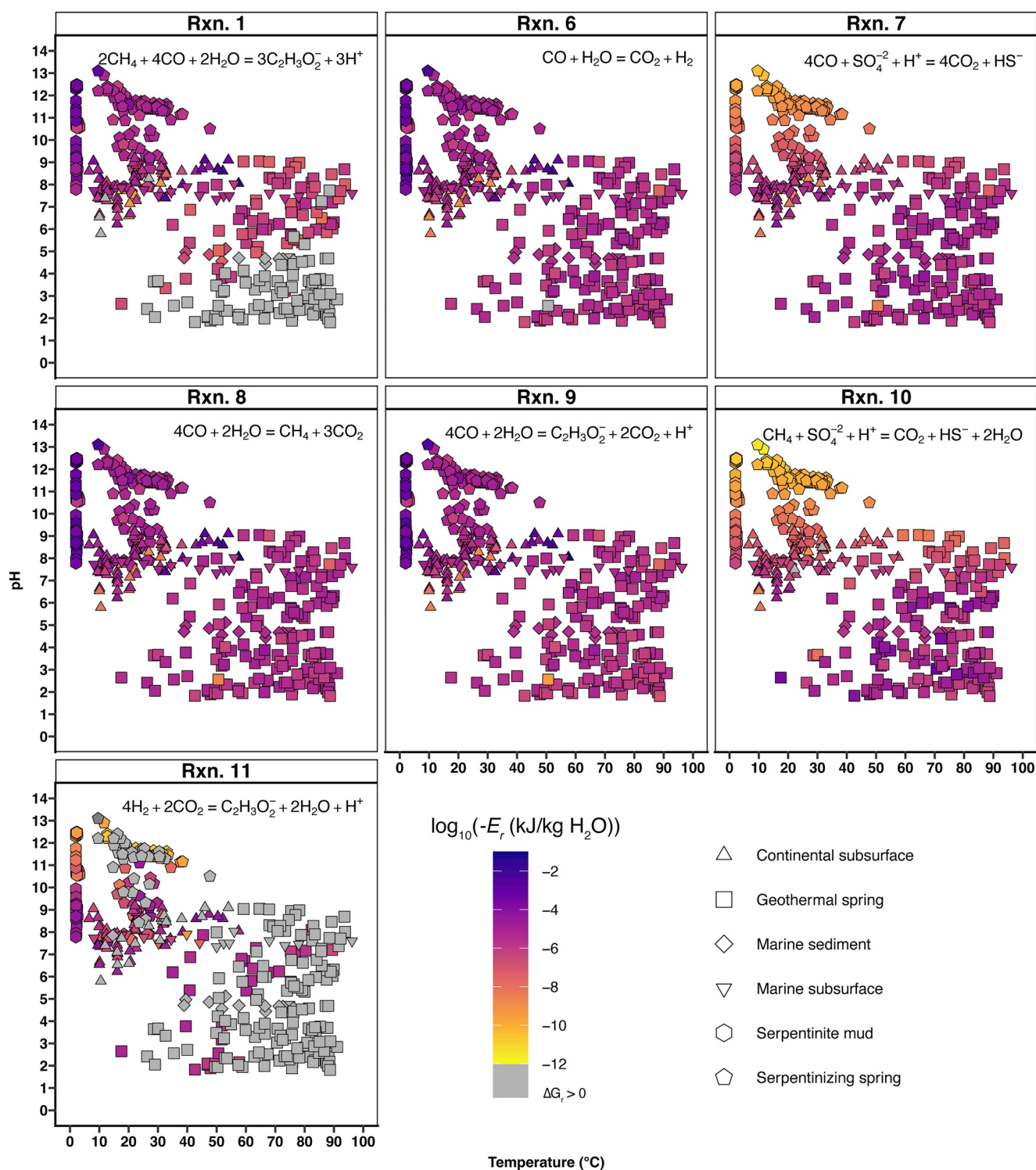


Figure 8. Energy densities (E_r , kJ/kg H_2O) for Reaction 1 and reference reactions (Reactions 6–11) across 439 natural samples. Energy densities represent the energy available per kilogram of water normalized by the stoichiometrically corrected activity of the limiting reactant. Temperature, pH, and activities of CH_4 , CO , acetate, sulfate, sulfide, CO_2 , and H_2 were sourced from speciated in situ geochemical data from seven environment types, represented as the shape of the points. Point fill indicates the $\log_{10}$ -transformed magnitude of E_r , where darker colors represent more higher energy densities (more exergonic values of ΔG_r); gray points denote samples where $\Delta G_r > 0$.

Table 1
Paired Comparisons of Energy Density Between Reaction 1 and Alternative Anaerobic Metabolic Reactions Across 439 Natural Samples

Reaction number	Reaction	<i>n</i>	Median $\Delta \log E$	Q25	Q75	Fraction of samples where Reaction 1 has a higher energy density	p_{adj}
6	Water-gas shift	344	+0.18	+0.07	+0.55	0.10	1.15×10^{-51}
7	CO oxidation via sulfate reduction	344	−0.66	−3.25	+0.37	0.62	5.16×10^{-17}
8	Methanogenic CO disproportionation	344	+0.26	+0.18	+0.63	0.003	2.60×10^{-57}
9	Acetogenic CO disproportionation	344	+0.19	+0.13	+0.52	0.003	2.60×10^{-57}
10	Methane oxidation via sulfate reduction	342	−1.74	−4.19	−0.37	0.80	3.35×10^{-33}
11	Homoacetogenesis	216	−1.17	−4.15	−0.21	0.79	7.81×10^{-20}

Note. Differences are reported as $\Delta \log E = \log_{10}(-E_r) - \log_{10}(-E_1)$, where negative values indicate higher energy density for Reaction 1. Comparisons are restricted to samples in which both reactions are exergonic (*n*). For each reaction, the table reports the number of paired samples (*n*), the median $\Delta \log E$ with interquartile range (Q25–Q75), and the fraction of samples in which Reaction 1 is more exergonic based on pairwise Gibbs energy comparisons. Reported *p*-values (p_{adj}) correspond to Holm-corrected Wilcoxon signed-rank tests of whether median $\Delta \log E$ differs significantly from zero.

density, where negative values indicate an advantage for Reaction 1. Reaction 1 yielded higher energy densities than CO oxidation coupled to sulfate reduction (Reaction 7), methane oxidation coupled to sulfate reduction (Reaction 10), and homoacetogenesis (Reaction 11) in 62%–80% of samples in which both reactions were exergonic, with median $\Delta \log E$ values of −0.66, −1.74, and −1.17, respectively, all differing significantly from zero (Wilcoxon signed-rank test, Holm-corrected $p_{\text{adj}} < 0.001$; Table 1). These values of $\Delta \log E$ correspond to median energy density advantages of 4.6-, 55-, and 15-fold for Reaction 1. In contrast, CO disproportionation reactions (Reactions 8 and 9) yielded higher energy densities than Reaction 1 in 99.7% of samples, and the water-gas shift reaction (Reaction 6) yielded higher energy densities than Reaction 1 in 90% of samples, with median $\Delta \log E$ values of +0.18, +0.26, and +0.19 for Reactions 6, 8, and 9, respectively, all differing significantly from zero ($p_{\text{adj}} < 0.001$; Table 1). These values of $\Delta \log E$ correspond to median energy density advantages of 1.5-, 1.8-, and 1.5-fold for Reactions 6, 8, and 9 over Reaction 1.

4. Discussion

Carbon comproportionation represents an underexplored class of redox reactions that may support microbial energy conservation. Here, we identify acetogenic CH_4 -CO comproportionation (Reaction 1) as a physiologically plausible metabolism and evaluate its energetics under natural and modeled environments. We assess the broader landscape of carbon comproportionation reactions and identify plausible candidates.

4.1. A Survey of Carbon Comproportionation Reactions

The distribution of nominal oxidation states of carbon (NOSC) provides a quantitative structure for exploring carbon comproportionation reactions. Although the standard state Gibbs energy of carbon oxidation (ΔG_{Cox}^0) can vary by >100 kJ/mol C among compounds sharing the same NOSC (Figure 1b), which may be attributed to differences in molecular structure including the presence of heteroatoms such as sulfur, nitrogen, and oxygen, the overall trend indicates that NOSC is a robust predictor of the energetics of carbon oxidation. Compounds with more negative NOSCs, and therefore more positive values of ΔG_{Cox}^0 , represent potential electron donors in comproportionation reactions, whereas compounds with higher NOSCs and more positive values of ΔG_{Cox}^0 represent potential electron acceptors.

However, most combinations represented are unlikely to support microbial catabolisms. Many of the most reduced carbon compounds occur as high-molecular weight hydrocarbons that are poorly soluble, degraded slowly in natural environments and infrequently serve as direct catabolic substrates for anaerobic microorganisms (Hu et al., 2025). For this reason, the reaction set in Figure 2 was restricted to small, labile, water-soluble compounds that are frequently involved in microbial carbon transformations. Even within this restricted set of reactions, standard state Gibbs energies alone do not identify physiologically plausible reactions. Negative values of ΔG_r^0 provide an initial indication of thermodynamic favorability, but reactions with positive values of ΔG_r^0 can become exergonic under certain environmental conditions. For example, elemental sulfur disproportionation

exhibits a standard state Gibbs energy of $+125 \text{ kJ mol}^{-1}$ at 25°C , yet still supports microbial growth because natural conditions can drive the overall Gibbs energy negative (Aronson et al., 2023). Accordingly, reactions with both negative and positive ΔG_r^0 values were considered in this initial screening, with the expectation that thermodynamic favorability under natural conditions may differ from standard state values.

By applying thermodynamic, geochemical, and biochemical considerations we can further refine the set of carbon comproportionation reactions that may serve as potential microbial catabolisms. Several acetylene-based reactions in Figure 2 yield negative ΔG_r^0 values; however, acetylene is infrequently quantified in natural anoxic environments, which limits the evaluation of these reactions across environmentally relevant conditions. As a result, acetylene-based comproportionation reactions are not considered further here. Independent of substrate availability constraints, many of the remaining thermodynamically plausible carbon comproportionation reactions represent simple reversals of fermentation reactions (e.g., $\text{CO}_2 + \text{ethanol} \rightarrow \text{glucose}$), and would require net ATP input, making them physiologically unrealistic despite favorable thermodynamics.

After applying these constraints, CH_4 oxidation coupled to CO reduction to form acetate (Reaction 1) emerged as a compelling candidate, hereafter “acetogenic CH_4 -CO comproportionation.” This reaction links two small, soluble, widely co-occurring carbon species and produces acetate, a central intermediate and common end product of anaerobic carbon metabolism, including acetogenic carbon fixation and fermentation. These attributes enable environment-specific evaluation of both Gibbs energy and energy density, allowing this reaction to be assessed not only in theory but also within the geochemical contexts where it may realistically occur.

4.2. Environmental Controls on the Energetics of Acetogenic CH_4 -CO Comproportionation

The energetic behavior of Reaction 1 reflects a combination of intrinsic reaction properties (ΔG_r^0) and strong influence of environmental chemistry. Across the temperature and pressure ranges examined, variations in ΔG_r^0 for Reaction 1 are modest, on the order of several kilojoules per mole of carbon, indicating that pressure exerts only a minor influence on reaction energetics relative to environmental chemistry (Figure 3). When ΔG_r is evaluated at environmentally realistic activities of CH_4 , CO, and acetate, Reaction 1 exhibits a strong dependence on pH (Figure 4). This dependence arises directly from reaction stoichiometry, as the net consumption of three protons strongly favors alkaline conditions; pH therefore exerts a first-order control on ΔG_r , while temperature plays a secondary role by modulating ΔG_r^0 . These controls produce systematic patterns across natural environments, with alkaline, reducing systems clustering in the most exergonic region of ΔG_r space (Figures 5 and 6a).

Among the environments examined, continental and marine serpentinizing systems consistently support the most favorable energetics for Reaction 1. These systems combine high pH, low to moderate temperatures, and highly reducing conditions that produce both CH_4 and CO. In serpentinizing systems, CH_4 is formed by CO_2 reduction with the abundant H_2 that is produced through the serpentinization reaction (Horita & Berndt, 1999; Leong & Shock, 2020; McCollom & Seewald, 2013). CO can also form abiotically in these systems through the reduction of CO_2 to form formic acid, followed by the decomposition of formic acid to CO (McCollom & Seewald, 2007; Seewald et al., 2005). In these settings, dissolved inorganic carbon is often sequestered as carbonate minerals, leaving fluids depleted in bioavailable $\text{CO}_2/\text{HCO}_3^-$ for autotrophic carbon fixation. Under such conditions, carbon species such as formate or CO may serve as important carbon sources (McCollom & Seewald, 2013), and, in the absence of oxidants such as oxygen, sulfate, nitrate, or ferric iron, may function as terminal electron acceptors.

Deep continental subsurface fluids frequently support favorable energetics for Reaction 1 across a broad range of conditions but exhibit some of the greatest variability in energy yields. These systems commonly maintain circumneutral to alkaline pH, low oxidant availability, and long-term isolation from surficial inputs, yet differ significantly in the availability of CH_4 and CO, which can be derived from abiotic reactions and thermogenic processes (G. Holland et al., 2013; Kieft et al., 2005; Sherwood Lollar et al., 1993, 2006). As a result, continental subsurface environments host some of the highest and lowest energy yields calculated for Reaction 1. Sites within the Witwatersrand Basin, for example, exhibit exceptionally high CO concentrations ($680 \mu\text{M}$ at Kloof Mine) and support some of the most favorable energetics, whereas fluids from the Deep Mine Microbial Observatory (DeMMO) contain extremely low (sub nanomolar) concentrations of both CO and CH_4 and yield correspondingly low or endergonic values of ΔG_r . This contrast highlights that, within otherwise similar subsurface geochemical regimes, variability in substrate availability—particularly CO—controls the energy yield of Reaction 1 and drives deviations from broader pH and temperature trends.

In contrast, geothermal springs and hydrothermal sediments exhibit highly variable energetics, which reflect their geochemical heterogeneity. Acidic, high-temperature systems tend to yield less favorable (more endergonic) values of ΔG_r , consistent with the dependence of Reaction 1 on high pH and lower temperature. Geothermal springs span some of the widest ranges of pH and temperature on Earth, from strongly acidic to alkaline fluids (pH 1.8 to 9.0) and from moderate to near-boiling temperatures (17.5–93.8°C), resulting in large ranges in predicted energy yields. Alkaline geothermal springs, however, yielded favorable energetics despite elevated temperatures, again highlighting the dominant role of pH over temperature in controlling ΔG_r (Figure 6). Marine sediments likewise exhibited broad variability in the energetics of Reaction 1. In hydrothermally influenced settings such as Guaymas Basin, thermolytic cracking of organic matter and magmatic inputs generate both CH₄ and CO (Conrad & Seiler, 1985; Lizarralde et al., 2023; Von Damm et al., 1985), while microbial methanogenesis contributes substantially to the CH₄ pool, which can exceed 100 mM at in situ pressures and temperatures (Bojanova et al., 2023). Despite high temperatures, these fluids supported exergonic conditions for Reaction 1 due to the high activities of reactants. In contrast, the hot and acidic shallow-sea hydrothermal sediments from Milos, Greece yielded the least favorable estimates of ΔG_r for Reaction 1, due to the combined low pH and high temperature (Figures 3, 4 and 6).

Across all the environments studied here, the most favorable conditions for Reaction 1 occur where high pH and persistent sources of both reactants co-occur. These patterns indicate that the energetic feasibility of Reaction 1 is largely constrained by the geochemical and geological contexts that sustain stable fluxes of CH₄ and CO over timescales sufficient for microbial utilization. Such environments provide the necessary foundation for evaluating whether and where Reaction 1 is a viable microbial catabolic strategy.

4.3. Comparison of Acetogenic CH₄–CO Comproportionation With Known Metabolic Reactions

The Gibbs energy of Reaction 1 was compared directly to six established anaerobic catabolisms involving CO, CH₄, and acetate (Reactions 6–11) across all 439 natural samples to place acetogenic CH₄–CO comproportionation within the thermodynamic context of known microbial metabolisms (Figure 7). When normalized per mole of carbon, Reaction 1 spans an energetic range that overlaps with these reactions, though the comparison reactions involving CO as a reactant (Reactions 6–9) are generally more exergonic under the same environmental conditions, reflecting the high thermodynamic favorability of CO oxidation. This highlights the limitations of evaluating energetic competitiveness solely on the basis of Gibbs energy yields.

Although Gibbs energies constrain the theoretical energy yield of a reaction, they do not reflect whether the required substrates are present at activities that allow that energy to be accessed by microorganisms. Normalizing Gibbs energies to the availability of the limiting reactant (i.e., expressing them as energy densities per unit mass or volume) provides a more ecologically relevant metric that better reflects which reactions can sustain microbial populations and shape biogeochemical processes in natural environments (LaRowe & Amend, 2014, 2019). Across the natural samples examined here, CH₄:CO activity ratios span more than eight orders of magnitude, causing Reaction 1 to be either CO-limited or CH₄-limited depending on the environment. Energy density calculations therefore reveal that the highest realizable energy yields for Reaction 1 occur in deep continental subsurface fluids and marine and continental serpentinizing systems, whereas environments with low CO or CH₄ activities yield orders-of-magnitude lower energy densities (Figure 8). Previous work applying energy density frameworks to continental subsurface and serpentinizing systems reported broadly similar ranges of accessible energy for established anaerobic metabolisms (Gure et al., 2026; Howells et al., 2025; Meyer-Dombard & Malas, 2022; Osburn et al., 2014).

To compare reactions based on accessible energy, paired differences in energy density were evaluated using the log-transformed metric $\Delta \log E$, calculated only for samples in which both reactions were exergonic. Reaction 1 exhibited higher energy densities than the water-gas shift reaction (Reaction 6) in ~10% of samples in which both reactions were exergonic (Table 1). Bacteria and archaea capable of catalyzing the water-gas shift reaction are typically thermophiles, with cultured representatives from anaerobic bioreactors (Alves et al., 2013; Parshina et al., 2010), hot springs (Balk et al., 2009; Novikov et al., 2011; Slepova et al., 2006, 2009; Sokolova et al., 2004; Svetlichny et al., 1991), and marine hydrothermal vents (Bae et al., 2006; T. G. Sokolova et al., 2001)—environments in which Reaction 1 tends to be least favorable. Additionally, in some facultative anaerobes, Reaction 6 may function only as a survival or maintenance strategy under electron acceptor limitation rather than a primary catabolic strategy (Katayama et al., 2026). The energy densities of Reaction 1 also exceed CO oxidation coupled to sulfate reduction (Reaction 7) in many high-pH, low temperature environments. In sulfate-reducers, CO

oxidation typically proceeds via the water-gas shift reaction (Reaction 6) with CO first converted to H₂ and CO₂, and the resulting H₂ powering sulfate reduction (Oelgeschläger & Rother, 2008; Parshina et al., 2005, 2010). This has been interpreted as a strategy to detoxify CO, as many sulfate reducers are inhibited by CO (Parshina et al., 2010). In many of the hyperalkaline sites (e.g., terrestrial serpentinizing) protons and/or sulfate limit the accessible energy yield of Reaction 7, while Reaction 1 remains energetically favorable.

CO disproportionation reactions (Reactions 8 and 9), yielded higher energy densities than Reaction 1 in the majority of samples (99.7%, Table 1). However, the energy densities of these reactions converge most closely with Reaction 1 in the marine subsurface, the continental subsurface, and serpentinizing environments, with median $\Delta \log E$ values ranging from +0.07 to +0.24 across these environment types, suggesting that direct competition for CO as a shared substrate is most likely in these environments. While CO disproportionation has been demonstrated in both methanogenic and acetogenic organisms, growth is often slow or inhibited at elevated CO concentrations (Daniels et al., 1977; Henstra & Stams, 2011; O'Brien et al., 1984; Rother & Metcalf, 2004). Methanogens capable of CO metabolism (e.g., *Methanobacterium thermoautotrophicum* and *Methanosarcina barkeri*) typically utilize Reaction 6 with the product H₂ used to fuel CO₂ reduction to methane (Daniels et al., 1977; O'Brien et al., 1984). In contrast, *Methanosarcina acetivorans* C2A lacks the ability to grow on H₂/CO₂, and instead sends CO-derived electrons into the Wood-Ljungdahl pathway to form acetate (Rother & Metcalf, 2004). Recent experimental work has extended these observations to anaerobic methanotrophic archaea, demonstrating that both freshwater and marine ANME archaea catalyze methanogenic and acetogenic CO disproportionation (Reactions 8 and 9, respectively), though these metabolisms may primarily support cell maintenance rather than growth at the CO concentrations tested, which substantially exceed those typically present in natural anoxic environments (Egas et al., 2026; Guo et al., 2026). Even where CO disproportionation is thermodynamically feasible, the underlying metabolic strategies and physiological constraints differ between organisms. The abundance, distribution, and ecological relevance of alternative CO-based metabolic strategies have been demonstrated in few pure cultures to-date, and remain poorly constrained. Further investigation in cultivars and in natural environments is needed to determine the ubiquity and importance of this process in nature.

The anaerobic oxidation of methane (AOM) coupled to sulfate reduction (Reaction 10) is an obligately syntrophic metabolism common in sulfate-rich marine sediments (Knittel & Boetius, 2009). AOM with sulfate yielded lower energy densities than Reaction 1 in ~80% of exergonic sites (Table 1), and was least favorable in some of the alkaline, sulfate-poor environments where Reaction 1 has the highest energy densities (Figure 8). Homoacetogenesis (Reaction 11) likewise yielded lower energy densities than Reaction 1 in ~79% of co-exergonic samples, though because homoacetogenesis does not compete directly for substrates with Reaction 1, acetogenic CH₄-CO comproportionation and homoacetogenesis could potentially coexist within the same environments. These comparisons show that when evaluated in terms of energy density rather than Gibbs energy alone, Reaction 1 is energetically competitive with or superior to CO oxidation coupled to sulfate reduction (Reaction 7), AOM (Reaction 10), and homoacetogenesis (Reaction 11), in the majority of co-exergonic samples, while CO disproportionation reactions (Reactions 8 and 9) and the water-gas shift reaction (Reaction 6) generally yielded higher energy densities across most environments (Table 1). Accounting explicitly for substrate availability and reaction stoichiometry provides a more realistic framework for assessing the energetic viability of carbon comproportionation reactions in nature.

4.4. Modeled Fluids From Mars and Early Earth

Modeled Martian and early Earth fluids allow the energetics of Reaction 1 to be constrained under inaccessible natural conditions using independently derived atmospheric and water-rock interaction models, in an approach analogous to that applied to natural terrestrial samples. Across all 186 modeled scenarios, including 164 early Earth ocean compositions and 22 Martian hydrothermal fluid models, ΔG_r values were negative, spanning −8 to −43 kJ/mol C.

Modeled Martian fluid compositions support the energetic feasibility of Reaction 1 across a wide range of plausible subsurface conditions. Using fluid chemistries derived from Marlow et al. (2014) and Rucker et al. (2023), which represent acidic sulfate alteration fluids, alkaline basalt alteration fluids, and serpentinization-associated analogs, Reaction 1 remained exergonic across all modeled scenarios evaluated. These fluids encompass a broad range of pH (0.67–11.86) and temperature (9°C–83°C), reflecting both the diversity of aqueous environments proposed for

ancient Mars and the uncertainty inherent in reconstructing Martian subsurface geochemistry. For each modeled fluid composition, aqueous CO concentrations were estimated from atmospheric mixing ratios of 1% and 50% using Henry's law (Koyama et al., 2024), while aqueous CH₄ activities were fixed at values adopted in the original fluid models (Marlow et al., 2014; Rucker et al., 2023). Across both low and high CO endmembers, predicted Gibbs energies overlap with those calculated for natural terrestrial subsurface environments (Figure 5). Although estimates of Martian geochemistry necessarily span a broad range of modeled fluid compositions—many of which have already been invoked to support other anaerobic methane metabolisms—Reaction 1 consistently fell within the thermodynamic range known to support microbial energy conservation.

For the early Earth, widely accepted models of atmospheric redox state assume a steady-state composition determined by mantle outgassing, in which CO₂ is the dominant gas, with only minor amounts of CO and virtually no CH₄ (Abelson, 1966; H. D. Holland, 1962, 1984; Poole, 1951). Multiple lines of theoretical and geochemical evidence, however, indicate that this steady-state framework does not capture the full range of Archean conditions. Large impacts, particularly during the late stages of accretion, are expected to have generated transient, and potentially longer-lived reducing atmospheres enriched in CO and CH₄ (Hashimoto et al., 2007; Schaefer & Fegley, 2007, 2010; Urey, 1952; Zahnle et al., 2010). An alternative interpretation holds that large impacts may not consistently generate strongly reducing atmospheres, as a substantial fraction of the reducing capacity of impactors could be sequestered in the crust and upper mantle rather than directly influencing surface waters and atmospheric chemistry (Citron & Stewart, 2022). Photochemical and impact-driven models have predicted CH₄ concentrations of 5×10^{-3} to 1.8×10^{-2} bar during portions of the Archean, (Catling & Zahnle, 2020; Catling et al., 2001; Kharecha et al., 2005; Ozaki et al., 2018; Wogan et al., 2023). CO abundances are less well constrained, but may have reached 10^{-5} to 10^{-2} bar during impact-driven reducing intervals, particularly when cometary CO ice or reduced carbon is delivered and oxidized (Catling & Zahnle, 2020; Schaefer & Fegley, 2007; Wogan et al., 2023). These ranges capture plausible post-impact and mantle-buffered endmembers during intervals following the last ocean-vaporizing impacts, when long-lived surface oceans and subsurface circulation would extend the residence times of CO and CH₄, sustaining redox chemistry on biologically relevant timescales. Depending on impactor size and composition, reducing conditions could persist for thousands to millions of years until hydrogen escape to space restored mantle-buffered redox conditions (Wogan et al., 2023; Zahnle et al., 2010). The early Earth ocean models therefore span both steady-state, mantle-buffered atmospheres with low CO and CH₄ and transient reducing periods in which these gases are substantially enriched. Across this range of atmospheric conditions, Reaction 1 remained exergonic, indicating that its energetic feasibility does not depend on transient or extreme atmospheric conditions. Moreover, if CO and CH₄ produced during transient atmospheric episodes dissolve into ocean waters and circulate at depth, their residence times may greatly exceed those of the atmosphere, particularly in subsurface environments isolated from rapid reoxidation. This decoupling of atmospheric composition and subsurface redox chemistry is consistent with the favorable energetics of Reaction 1 observed in modern serpentinizing systems and deep subsurface fluids.

The highest energy densities for Reaction 1 occurred in deep continental subsurface fluids and serpentinizing systems, which are characterized by isolation from phototrophic energy sources, low energy fluxes, and redox disequilibria sustained by water-rock reactions. These conditions define a class of energy-limited subsurface environments that are increasingly central to astrobiological investigation. In this context, Earth's subsurface environments provide relevant analogs for extraterrestrial environments because of shared energetic and geochemical constraints. Serpentinization represents a key process analog linking these systems. Serpentinization-hosted alkaline hydrothermal systems have also been widely invoked as plausible settings for early metabolism on Earth (Martin et al., 2008; Russell et al., 2010), particularly in models that place acetogenesis and the Wood-Ljungdahl pathway at the core of some of the earliest energy-conserving reactions (Russell & Martin, 2004; Weiss et al., 2016). Within this framework, reactions coupling CO and CH₄ to acetate formation analogous to Reaction 1 have been discussed as plausible components of early acetogenic metabolism at alkaline hydrothermal vents (Nitschke & Russell, 2013; Russell & Nitschke, 2017). Consistent with this view, deeply branching acetogenic lineages have been identified in serpentinite-associated subsurface fracture waters, including members of the candidate phyla *Acetothermia*, *Lithacetigenota*, and NPL-UPL2, as well as the acetogenic archaeon within the order *Methanocellales* (Colman et al., 2022; Nobu et al., 2023; Suzuki et al., 2018, 2024). While these organisms do not utilize direct CO-CH₄ reactions, they employ canonical or modified Wood-Ljungdahl pathways, in some cases using CO, formate, or glycine as substrates for acetogenesis. These

observations highlight the long-standing association between serpentinization-driven environments, C1 chemistry, and acetogenic energy metabolism.

Exploration targets for life detection—including Mars, Europa, and Enceladus—possess surface environments that are largely inhospitable to life but are thought to host, or to have hosted, subsurface aqueous systems where chemical energy governs habitability (Jones et al., 2018; Michalski et al., 2013, 2018). Evidence for serpentinization has emerged on other planetary bodies. Detection of H₂ in the plume of Enceladus (Waite et al., 2017) has been attributed to ongoing water-rock serpentinization reactions in an alkaline subsurface ocean (Glein et al., 2015; Glein & Truong, 2025), while models of Europa's ocean-seafloor interface likewise predict water-rock reactions capable of generating redox disequilibria, even where fluid pH remains poorly constrained (Hand et al., 2007; McCollom, 1999; Vance & Melwani Daswani, 2020; Vance et al., 2007, 2016; Zolotov & Shock, 2003, 2004). On Mars, the occurrence of olivine and serpentine minerals in the Noachian-aged crust indicates past aqueous alteration consistent with subsurface serpentinization (Ehlmann et al., 2010; Hoefen et al., 2003; Ody et al., 2012). Taken together, these observations suggest that the geochemical conditions that favor Reaction 1 may be present in the subsurface of other planetary bodies.

5. Conclusion

In this study, we identify carbon comproportionation as a previously underexplored class of anaerobic redox reactions with the potential to support microbial energy conservation in low-energy environments. We show that acetogenic CH₄-CO comproportionation is exergonic across a broad range of temperatures, pH conditions, and natural fluid compositions, and furthermore that energy yield depends strongly on substrate availability rather than the Gibbs energy alone. By explicitly incorporating the availabilities of limiting reactants in each environment through energy density calculations, we demonstrate that this reaction occupies an energetic range comparable to and in some settings exceeding that of established anaerobic metabolisms involving CH₄, CO, and acetate. The highest energy densities occur in subsurface and serpentinization-associated environments, reflecting the role of water-rock reactions in producing and maintaining CH₄ and CO under reducing and often alkaline conditions. These results collectively expand the known range of energetically viable anaerobic carbon transformations, providing a quantitative framework for evaluating carbon-based catabolic reactions under realistic geochemical constraints, and motivate further investigation of carbon comproportionation reactions as potential contributors to subsurface biogeochemical cycling on Earth, and by extension, on other planetary bodies.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All data are available in Supporting Information S2. Data and R scripts written for data analysis and visualization are available (Aronson, 2026).

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